NAVIGATION / +C THROUGH +C4

Quick reference card

From Pads, select a pad and press **INFO twice** to open its parameters. On a supported parameter, **select + INFO** opens modulation. Page numbers refer to the full reference guide.

Waveform · 2	Pad > INFO. Spread or pinch vertically for 25-3200% amplitude, horizontally for time. A still two-finger tap restores 100% amplitude and horizontal fit.
Stop / reset · 2	On Pads, hold a **Trigger** pad for 3 s to stop it after its usual retrigger. Double-tap a numeric dial to reset its base value and retain its routes.
Speed / Pitch · 3	**Sample > LFO/Spd > SPEED**, 50-200%, one eligible voice. Pitch works across pads and on the Speed voice. SPDBUSY / RETRY concern Speed only.
Tone · 4	**Pad > Tone**. Fidelity, Drive, Filter 2, Res 2 and Slope 1. Select + INFO on Drive, Filter 2 or Res 2 for modulation.
Clip / Slicer · 5-6	**Clip > Pos** for Loop Fade, Shift, Play Len, Sync, Live Flip and Fill every. **Slicer > Pos** for Sync, Sync Grid and **Slice Fit > Cut / Tape**.
GEN · 7-8	**Pad > Gen** for algorithm, steps and timing. **Cnf** for Cond, Proc and Resp Pad. Granular has all GEN controls in Cnf. **Tools > Mstr > Section > GEN Run > Run On** starts the generators.
Delay · 9	**FX1 > INFO > More**. Flutter, Mod Rate, Mod Depth, Phase, Drive, Tone and Return dB. Codec shows Loss and Bandwidth.
Reverb / FX3 · 10	**FX2 > INFO** opens reverb. Tap the selected FX2 cell again for **FX 3**, then INFO. More opens detailed Room/Hall/Chamber controls.
Sends / outputs · 9-11	**Pad > Pos > FX1/2/3 Send**, then INFO. **Tools > Rec > FX Out** routes FX1/2. FX3 Out is on its Plate page or More for the other types.
Modulation · 11-12	**Source** includes KEY, ENV, CHAOS, RAND and NET. Up to 12 saved routes per pad. Slicer Slice Seq responds to source edges.
CV / MIDI · 13, 16	**Tools > Rec > CV In** for Scale, Offset, Quantize and Root. **Tools > MIDI > Monitor** for incoming messages.
Master · 14-15	**Tools > Comp** for compression. **Mstr > Section** for Comp env, Master FX (shaper / Drive / Sweep), Dist Ctrl (Tone / Mix / Level) and Sweep Mod.
Display / reset · 2, 14	**Tools > Disp** for saver, signal wake and line graphic. Double-tap the counter to renumber the next bar. **Tools > Main > RESET** offers Tools or FX.
Recording / update · 16-17	Record dry while monitoring through the pad sends. For installation or rollback, put **BITBOXMK2.bin** in the card root and power up holding INFO.

+C WAVEFORM AND SAVER / +C3 DISPLAY / +C4 HOLD TO STOP

Waveform, pads and display

Pad gestures and global display controls

Waveform and numeric dials

Press INFO on a loaded pad. Spread or pinch vertically for **25-3200% amplitude**, horizontally for time, or diagonally for both. A still **two-finger tap** restores 100% amplitude and horizontal fit. Zoom affects only the view.

Double-tap a numeric dial on a pad or FX grid to restore its base value. Pitch returns to zero, for example. Its saved modulation can still move the effective value.

Stop a Trigger pad [+c4]

On the 4 × 4 Pads grid, a **Trigger** pad plays or retriggers on contact. Hold the same pad for **three seconds** to release all its voices once. Other pads and effect tails can continue. GEN, MIDI, CV or a new tap can trigger it again. Slide off, add a finger or leave Pads to cancel the hold.

Tools > Disp

Scrn saver	Choose **disabled** (default) or **1, 2, 3, 5, 10, 15, 30 or 60 minutes**. Playback continues while the screen sleeps. The first touch, button or encoder movement only wakes it.
Sig wake	Choose disabled, MIDI, CV trig or both. Qualifying signals wake the display, restart its idle timer and still play normally. MIDI Clock and Active Sensing are ignored. CV wake uses a rising trigger or gate.
Saver gfx [+c3]	Off gives a black screen. On draws ten fine lines in shades of blue, purple and white at the current brightness. One line changes every half-second.

Renumber the counter [+c3]

Double-tap the top-left bar:beat counter. A star marks the pending reset. The next bar reads **1:1** and playback, clock phase and periodic fills keep their timing. The new display origin lasts for the session.

Display preferences save globally. The stock **MIDI In** tab is now labelled **MIDI**.

SPEED +C / LIVE RETRY +C3 / PITCH COEXISTENCE +C4

Speed and native Pitch

Sample parameters > LFO/Spd > SPEED

Set duration and pitch independently

Drag SPEED or turn its encoder. **50%** plays the Sample in twice the time and **200%** plays it in half the time. Speed preserves the tuning set by Pitch. At 100% with zero Speed modulation, the voice uses native playback. Select SPEED and press INFO to assign modulation.

Native **Pitch works on every playing pad, including the Sample voice using Speed**. For example, set a 48 kHz Sample to +24 semitones, then compare 50%, 100% and 200% Speed. The rhythm changes while the sample stays two octaves up, within the native Pitch range. Higher file rates and MIDI transposition count toward the original pitch ceiling.

One active Speed voice

First request	The first eligible forward Sample voice that requests Speed gets the slot. A live adjustment on a playing Sample can acquire a free slot. Polyphonic notes on the same pad also share that single slot.
SPDBUSY	A waiting voice plays at its own Pitch and 100% Speed. You can still edit its saved Speed and routes.
RETRY	Once the owner stops or releases the slot, a waiting voice continues at 100%. Restart it, or return Speed to 100%, let that take effect, then adjust again.
Release live	Set the owner to 100% and its Speed modulation depths to zero. The slot becomes available after the short handoff to native playback. A nonzero Speed route holds the slot through crossings of 100%.

Speed and its routes save with the pad. Older presets start at 100%. Reverse playback, loop crossfade, modulated loop boundaries and unsupported source configurations use the native playback path.

CPU load: one stretched Sample plus several pitched voices and effects can produce crackling. A hardware test with effects active reached that point when a third pad joined. The practical limit depends on the preset. Stop a voice or bypass an effect if needed.

TONE +C2 / CODEC FIDELITY MODES +C3

Tone and Fidelity

Pad parameters > Tone, on Sample, Clip, Slicer and Granular pads.

Fidelity replaces Interp [+c2]

Normal and **HighQ** retain the native choices. **SP1200, SP-12, S950, MPC60 and SP Raw** add sampler-inspired coloration. Fixed-rate choices combine 12-bit quantization with **16, 12, 10, 8, 6, 4 or 2 kHz** rate reduction.

Phone [+c3]	8 kHz mu-law coloration with telephone-band low-pass filtering.
ADPCM [+c3]	IMA-style adaptive 4-bit quantization at 22.05 kHz.
u-Law [+c3]	Mu-law coloration at 16 kHz. The module uses the ASCII label u-Law.

The new modes follow 2k, preserving earlier saved selection numbers. Stereo channels have independent codec histories. Fidelity remains a manual selector.

Drive and two filters [+c2]

Drive	0-100% soft saturation before the filters; default 0%. Select + INFO for modulation.
Slope 1	12 or 24 dB/octave for the original Main-page Filter; default 12 dB. Move Filter away from center to hear the difference. Manual selector.
Filter 2 / Res 2	Independent 24 dB/octave filter. Negative selects low-pass, positive high-pass, center bypasses it. Res 2 defaults to 50%. Both support select + INFO modulation.

The left bank has Fidelity and Drive above Filter 2 and Res 2. Slope 1 sits at the top of the right bank. Tap a bank to give its controls the encoders.

Double-tap resets numeric base values while keeping routes. Returning from modulation restores the same Tone dial and encoder bank. The configuration tab is shortened to **Cnf**. Blank pads expose **LFO/Spd** and Speed at 100%.

CLIP TOOLS +C2 / CLOCKED ADDITIONS +C3

Clip timing, fills and slices

Pad parameters > Pos. Scroll the list to reach the added controls.

Loop Fade [+c2]	0-50 ms, default off. Dips loop edges toward zero to reduce clicks. Capped at a quarter of a short loop; try 2-5 ms initially.
Shift [+c2]	-100 to +100 ms. Positive starts later in the file; negative brings its tail forward. Wraps at the source boundary.
Play Len [+c2]	Full or 1/4-32 bars in quarter-bar steps. Caps at the available source, rounding to the nearest Sync division with a one-division minimum.
Sync 1/32 [+c2]	Adds a finer timed Sync division. Play Len, Live Flip and Fill every require Quant Size enabled and a timed Sync division; not Slice sync or None.

Live Flip [+c2], Breakcut and periodic fills [+c3]

Beat Rpt plays beats 1,2,2,2; **Half Rpt** plays 1,2,1,2; **Rotate** plays 2,3,4,1. **1/8 Rpt** and **1/16 Rpt** repeat subdivisions in beat four. **Breakcut** adds changing sixteenth-note cuts/repeats while keeping the downbeat.

Fill every: Always / 2 / 4 / 8 / 16 bars. At 4 bars, for example, bars 1-3 play normally and bar 4 uses the selected Live Flip. Fills follow the underlying musical clock, including short loops. Counter renumbering does not change their phase. Off rejoins the current source position.

These Clip effects run independently of Sample Speed and GEN. Set Beat Count to match the loop. Cuts through sustained sound can click.

Slice Sync Grid [+c3]

For a Slicer pad with **Sync On**, choose Free / 1 bar / 1/2 / 1/4 / 1/8 / 1/16 / 1/32. Synchronized slice duration rounds to a whole grid unit, minimum one. Free uses the native synced duration. **Slice Fit** sets how the Slicer plays within that duration. See page 6. Sync Off uses native playback.

Slice Mono fix [+c2]: after switching a polyphonic Sample pad to Slicer, the saved Mono setting is honored. With Trigger, Mono and Play Through On, a new slice cuts the old one even with Quant Size None.

SLICE FIT / +C4

Slicer Tape

Slicer pad > INFO > INFO > Pos > Sync On > Slice Fit

Fit the whole slice to its duration

Select **Slice Fit** immediately below Sync Grid. **Cut** is the default for new and existing presets. In Cut, the native reader plays at the pad's pitch rate, so a synced duration can trim the end or leave a gap. **Tape** plays from one marker to the next at the rate required to fill the synced duration.

Longer duration	A 125 ms slice fitted into 250 ms plays at half speed, one octave lower.
Shorter duration	The same slice fitted into 62.5 ms plays at double speed, one octave higher.
Pitch	Tape derives the fitted slice's pitch from its duration. Use Cut for ordinary Slicer Pitch control. Tape runs alongside the single Sample Speed voice.
When edits apply	Set the mode or change the grid, then launch a fresh slice to hear the new fit. Sync Off uses the native playback rate.
Limits	Very short target durations that cannot contain the native fades, invalid markers and extreme read rates fall back to Cut.

Audition a break

Choose a break with clear kick and snare markers. Enable Sync, choose a grid division and trigger the same note in Cut and Tape at a slower tempo. Listen to the pitch and to the last transient before the next marker. Then try Play Thru or Slice Seq.

Sample Speed preserves pitch while changing duration on an eligible Sample voice. **Slicer Tape** ties rate and pitch together on synced slices. Slice Fit saves with the pad.

PER-PAD GENERATORS / +C4

Start a GEN pattern

One clocked generator per pad, up to sixteen running together.

Build a kick-and-hat pattern

1. Load a short kick on a Sample pad. Press INFO twice and open **Gen**. Select **Euclid**, Steps 16, Hits 4 and Rate 1/16. Set another pad with a hi-hat to Euclid, Steps 16 and Hits 5.

2. In **Tools > Mstr > Section > GEN Run**, set **Run On**. The patterns follow the Bitbox musical clock, including incoming MIDI clock. Return to the hat and turn Rotate one step to move it against the kick.

3. Lower Prob to skip occasional hits. Gen Vel sets the generated velocity. Notes sound on the scheduled step. Gate pads release at the programmed gate end. Set Gen to Off to silence one generator. GEN Run > Run Off stops scheduling on all pads and releases their gates. Triggered one-shots and effect tails can finish.

Find the controls

Pad > Gen	Sample, Clip, Slicer and Multisample have eight dials: Gen, Hits, Steps, Rotate, Rate, Prob, Var and Gen Vel. Touch a bank to assign its four encoders.
Pad > Cnf	Gen Cond selects a condition, Gen Proc selects a Breakcut procedure, and Resp Pad chooses the Call+Resp answer pad. Granular keeps its Gran tab and places all eleven GEN controls at the end of Cnf.
Tools > Mstr > Section	GEN Run: Run, Fill, Seed. GEN Play: Epoch, Turn, Mutate. GEN Cut: Cut, Rolls, Recut. GEN Learn: Learn, Scale, Range.

Pad GEN settings save with the preset. Global GEN settings save under Tools. Defaults are Run Off and Gen Off on each pad. If Run is already on, loading a preset can start its configured generators.

PERFORMANCE REFERENCE / +C4

GEN algorithms and status

Var has a different musical function for each algorithm.

Euclid	Distributes Hits across Steps. Rotate shifts the pattern. Var above its midpoint introduces soft ghost hits.
Breakcut	Rearranges break or slice segments. Choose Gen Proc under Cnf, then enable Cut under GEN Cut. Var changes how much is rearranged.
Call+Resp	Three call cycles lead to an answer on the fourth. Choose Resp Pad to send the answer to another pad. Var controls response density.
Evolve	Changes the rhythm over successive cycles. Var controls mutation.
Drummap	Pad columns take kick, snare, hat and percussion roles. Rotate and Var move across the drum map.
Random	Hits controls average density. Var spreads velocities. This is a rhythm generator; the RAND modulation source is described on page 12.
Acid	Generates notes using the global Scale and Range. Var influences accents and slides.
Turing	Generates a repeating pitched pattern. Turn Var toward 100 to lock it.
Learn	Select Learn on the pad and turn on Tools > Mstr > GEN Learn > Learn. Tap the pad to teach timing gaps. Var controls adherence. Learned history lasts for the powered session.

Read the status strip



With Run on, select a pad whose generator is enabled. **Tall white bars** show programmed hits. **Short gray bars** show rests. The **yellow playhead** stays above the current step. Conditions and probability can suppress a programmed hit.

Cyan dashes below the pattern identify enabled generators. A dash flashes white on an emitted note and a yellow underline marks the selected pad. Selecting a pad with Gen Off clears the strip. Recording progress and longer native messages take priority.

DELAY AND ROUTING +C3 / EXPANDED CONTROLS +C4

Delay and FX return routing

FX1 > INFO for delay. Tools > Rec > FX Out for the physical returns.

Choose **Digital** for the original delay, **Tape** for driven repeats with drift and flutter, **BBD** for darker repeats that grow coarser at longer delay times, or **Codec** for accumulating 4-bit ADPCM coloration. Delay, Feedback, Cutoff, Width, Beat Sync, Filter and Ping retain their native controls.

FX1 > INFO > More

Flutter	Adds 0-100% flutter depth on all four types, default 0. Tape also has built-in flutter.
Mod Rate / Depth	0.05-10 Hz wow rate with up to 10 ms of movement. Depth defaults to 0. Raise Depth to hear Rate.
Phase	Stereo modulation offset from 0 to 180 degrees, initially 90.
Drive / Tone	Drive saturates sound entering the delay line. Tone sets repeat bandwidth. Effects accumulate on successive echoes. Start at Drive 0% and Tone fully open.
Loss / Bandwidth	In Codec mode, Drive becomes Loss (0-30% chance of dropping a whole 10 ms frame). Tone becomes Bandwidth (4-16 kHz codec sample rate). Defaults are 0% loss and 16 kHz.
RETURN DB	Wet return gain from -12 to +12 dB, default 0 dB. This changes echo level on its chosen output independently of Feedback.

More and Main switch pages. Delay settings save with the preset. Double-tap a numeric dial to restore its default.

Tools > Rec > FX Out

Out1/2	Default. FX1 delay and FX2 reverb return to the main stereo pair.
Dly FX1/2 / Rvb FX1/2	Moves the named effect to the physical FX stereo pair. The other effect stays on main.
FX1/2 / Dly1 Rvb2	FX1/2 moves both effects to the FX pair. Dly1 Rvb2 sends delay to mono FX1 and reverb to mono FX2, folding each stereo return to mono.

FX Out is global. FX3 has its own output setting. Master distortion and Sweep process Out1/2, while compressor routing follows the selected bus.

REVERB TYPES +C2 / MORE CONTROLS AND FX3 +C3

Reverb and the third send

FX2 > INFO, or select FX3 using the pad-grid toggle.

Reverb types and main controls [+c2]

Plate is the original algorithm and default. **Room, Hall and Chamber** add Size, Diffusion and Width. All types offer Level: 50% is unity, with about -12 to +12 dB available. Decay, Predelay and Damping retain native modulation. Type changes briefly fade and clear the previous tail.

Room / Hall / Chamber > More [+c3]

Bass x / Xover	Bass decay from 0.5x to 2x relative to midrange; center 1x. Crossover approximately 100 Hz-2 kHz.
Mod Rate / Dpth	Internal reverb modulation, approximately 0.1-5 Hz, with adjustable depth.
Early	Early-reflection level.
Hi Cut / Lo Cut	Input high-frequency cutoff approximately 1-14 kHz; low-frequency cutoff approximately 20-500 Hz.
Level / Main	FX2 repeats its output Level here; FX3 uses this square for FX3 Out and keeps Level on its main page. Main returns without clearing the tail.

FX3: a second independent reverb [+c3]

On Pads, tap FX2 once to select it, then tap the selected cell again. Its label becomes **FX 3**. Press INFO for parameters. Another tap on the selected cell returns it to FX2. FX3 has the same four algorithms and detailed controls, with its own settings and tail.

FX3 Out selects Out1/2 / FX1/2 / FX1 / FX2 independently. It is on Plate's main page, or under More for Room/Hall/Chamber. Mono routing averages both wet channels. An enabled compressor processes the destination pair; master distortion and Sweep remain on main only.

FX3 Send is under each pad's **Pos** page and defaults to zero. FX2 and FX3 settings save independently with the preset. The grid toggle changes only which effect is being viewed.

FX3 skips processing while unused and lets a tail run after its last input. Use representative performance loads when checking headroom, especially Speed, two reverbs and compression on Both.

THIRD CUSTOM RELEASE / +C3

Send modulation and KEY

Pad > Pos > FX1 Send / FX2 Send / FX3 Send > select > INFO

Move the sends with modulation

The usual source/amount editor now works for all three pad sends, in every pad mode, including empty pads. LFO, external CV, velocity, KEY, MIDI CC and other native sources can control the send. HOME returns to Pos and the same selected row.

Modulation adds to the manual send level, then clamps to 0-100% before the native send curve. A zero manual send can therefore open under positive modulation. Remove the route to restore the manual level. The original mixer controls set those same underlying levels.

Each send offers three route slots, within the native **12 saved modulation routes per pad total** across all destinations. This limit also includes Pitch, Speed, filters and other assignments.

What happens with polyphony?

A send processes the pad's summed audio. For voice-dependent **LFO, VEL and KEY**, the last active voice in the native list supplies that pad's send modulation; Clip/Slicer uses its last active embedded voice. This is not an independent send envelope for every note and does not promise newest-note priority.

External CV and controller sources are shared independently of that voice selection. When no voice is active, voice-dependent sources contribute zero. Send settings and routes save with the preset; recording retains the send routes too.

KEY exposes native note tracking

Choose **KEY** after CC in the modulation source selector. It uses the engine's existing note-position source, centered on MIDI note 60. Positive or negative amounts can open or close a filter as you move up the keyboard, for example.

KEY is available on supported native destinations and custom Tone, Speed and send destinations. Per-voice parameters retain per-voice tracking; sends use the pad rule above. KEY differs from **PTCH**, the existing pitch-controller source. Older source-selector indices are unchanged.

PER-PAD MODULATION / +C4

Four pad modulation sources

Select a supported pad parameter, press INFO, choose Source and set its Amount.

After KEY, the source list offers **ENV, CHAOS, RAND and NET**. Each of the 16 pads has independent state for all four. Two routes from one source on the same pad share its current value. Use positive or negative Amount to set the direction and depth.

ENV	A single envelope for the pad follows its Attack, Decay, Sustain and Release settings. Each admitted voice start retriggers the attack from the current level. The source runs from zero to one.
CHAOS	A continuously evolving, fixed-rate bipolar signal generated by a Lorenz attractor. Use a small amount on Filter 2 for irregular movement.
RAND	A new bipolar value on each admitted note, held until the next hit. Repeated triggers on a mono pad also advance it. Try RAND on Pan or FX1 Send.
NET	Four coupled nonlinear nodes produce slow, evolving bipolar movement at a fixed rate. Try NET on Drive or a reverb send.

Slicer Slice Seq uses edges

ENV and RAND advance Slice Seq on each admitted note. CHAOS and NET advance it when their signal crosses zero in the rising direction. The source supplies a trigger count, so changing Amount does not scale the number of steps.

Routes and recall

The pad’s source, destination and amount save with the preset. Envelope levels, random values and the live phases are playback state. Clearing the last route retires that source. Available destinations depend on the pad mode.

For one changing hit at a time, use **RAND**. For slow movement through a sustained note or effect tail, try **CHAOS** or **NET**. Master Sweep has a separate global source selector on page 15.

THIRD CUSTOM RELEASE / +C3

Shared CV utilities

Tools > Rec > CV In. Back returns to the ordinary Rec controls.

Scale and offset external modulation

CV Scale %	0-400%, default 100%.
CV Offset %	-100 to +100%, default 0%. Added after scaling.
Shared scope	Applies to all eight EXT modulation sources together, not separately per socket. IN1/IN2 audio sources are excluded.

At Scale 200% and Offset -50%, a normalized input value of 0.6 becomes 0.7. The transformed modulation value is limited to -2 to +2; each destination keeps its own range limits. Editing these settings also updates a held voltage without needing the CV to move.

Quantize CV-controlled pitch

Quantize offers Off, Major, Minor, Dorian, Phrygian, Lydian, Mixolydian, Harmonic minor, Major pentatonic, Minor pentatonic, Blues and Whole tone. **Root** selects C through B. Defaults are Off and C.

The native CV-pitch source snaps to the nearest allowed note; ties select the lower note. It is **CV-only**: it does not quantize MIDI notes or the final sum of unrelated pitch modulations. Scale/Offset affect the general EXT modulation values; pitch quantizing operates separately on the native CV-pitch source.

Triggers, saving and reset

Raw CV still reaches the native gate/trigger detector first. Trigger thresholds, rising edges and signal wake remain unchanged. A scale or root edit does not synthesize another trigger.

All four settings save globally through the normal Tools store. **RESET > Tools** restores 100% / 0% / Off / C. RESET > FX leaves them alone. Leaving Rec closes the subpage, so returning to Rec shows its ordinary controls.

For an initial check, keep Scale 100%, Offset 0% and Quantize Off, then change one setting at a time. The default settings preserve native CV behavior.

MASTER CONTROLS +C2 / ROUTING AND RESET +C3

Master processing and RESET

Tools > Comp and Tools > Mstr > Section

Compressor controls [+c2]

Comp page	Threshold -48 to 0 dB; Ratio 1:1 to 20:1; Gain -12 to +24 dB; parallel Mix 0% dry to 100% processed.
Comp env	Tools > Mstr > Section > Comp env: Attack 1-200 ms and Release 10-2000 ms. Defaults: 10 ms and 250 ms.
Defaults	Threshold -4 dB, Ratio 4:1, Gain -4 dB and Mix 100%. The original compressor algorithm remains in use.

Compressor bus [+c3]

Off / Out1/2 / FX1/2 / Both appears under Tools > Main and Mstr > Comp env. These are two views of **the same setting**. FX1/2 means the physical FX output pair. Both uses independent histories/detectors with shared controls. Mono assignments join their output pair before compression.

Master FX [+c2]

Mstr > Section > Master FX offers **Soft, Hard, Fold or Tube** distortion, or Off; Drive adds 0 to +24 dB. Sweep left for low-pass, right for high-pass, center for dry. On Out1/2 the order is distortion, Sweep, then optional compression. These settings save globally; changing compressor buses does not move distortion or Sweep.

RESET with two scopes [+c3]

Tap **Tools > Main > RESET**, beside the version. Choose Tools or FX, then **Reset** to confirm. Cancel or leave the panel to dismiss it.

Tools	Resets native/custom global settings: display, CV, MIDI/clock, recording options, routing, compressor and master processing. Saves one global-settings snapshot.
FX	Resets FX1, FX2 and FX3 scalar settings. Changes the current preset; save the preset to retain the reset. Preserves Tools values.

Both scopes preserve pads, samples, pad parameters, send levels and modulation routes. Neither deletes files or overwrites a saved preset.

EXPANDED MASTER PROCESSING / +C4

Distortion blend and Sweep

Tools > Mstr > Section. Process the main Out1/2 stereo mix.

Blend distortion in parallel

In **Master FX**, select Soft, Hard, Fold or Tube, then set Drive (0 to +24 dB). Switch Section to **Dist Ctrl** for Tone, Dist Mix and Level dB.

Tone	Filters the distorted signal before the blend. Lower it to darken the grit while keeping the clean component bright. Fully open is the default.
Dist Mix	0% clean through 100% distorted, default 100%. The clean and distorted paths are time-aligned.
Level dB	Trims the combined signal from -24 to +12 dB while a shaper is enabled. Default 0 dB. Use it to match loudness after raising Drive.

Modulate the master Sweep

Use the **Sweep** control on the Master FX page to choose the center position. Left is low-pass, center is dry, and right is high-pass. In **Sweep Mod**, choose a source and raise Depth.

Source	Off, EXT1-EXT8 or LFO. EXT uses the corresponding CV jack with the global CV Scale and Offset.
Depth	Signed amount. A negative setting reverses the direction. Defaults are Source Off and Depth 0%.
LFO Rate	One triangle cycle per sixteenth, eighth, quarter or half note, or per 1, 2, 4 or 8 bars. The default is 1 bar. The LFO follows the musical clock independently of GEN Run.

Try a drum loop with Sweep at -40%, Source LFO, Depth +30% and Rate 1 bar. Distortion feeds Sweep, followed by compression when enabled. These master settings save with Tools.

Recording and MIDI monitor

Monitor a dry take through effects, or inspect MIDI arriving at the module.

Recording keeps the sends

[+c2] FX1/FX2 send levels survive recording, and the selected pad's sends process live input monitoring. **[+c3]** This includes FX3 and the modulation routes on all three sends. External CV/controller modulation can work while the pad has no active voice.

Monitor On / Auto / Off keeps its native meaning. Selecting another pad changes which send settings monitor uses. The WAV contains the input, including input gain; the effects heard through sends stay out of the recorded sample.

Tools > MIDI > Monitor [+c2]

Tap Monitor beside the version number. Capture starts on entry. Seven visible rows show the newest entries from a 32-event history. Each row gives elapsed time, input group, channel and message. **0.350 A 03 CC 74=96**, for example, means CC 74 at value 96 on channel 3 after 350 ms.

Freeze / Resume	Freeze holds the displayed history while capture and playback continue. Encoder 2 scrolls frozen rows. Resume returns to the latest events.
Channel filter	Touch Ch All / Ch 1-16 or use encoder 1. Transport messages appear with every channel setting.
Clear / Back	Clear restarts elapsed time and clears the log. Back stops capture and returns to MIDI settings.
CLK / SENSE	Separate activity lights for MIDI Clock and Active Sensing. Rows show notes, CC, program changes, pressure, pitch bend, transport and reset messages.

A/B are native input groups. B includes the USB queue and one parser path, so a group is not a unique jack identifier. SysEx and system-common messages are discarded before this monitor. Signal Wake is configured separately under Disp.

Playback state [+c3 / +c4]

The +c3 memory refactor moved the dial-reset workspace into reserved SDRAM and compacted playback state. +c4 uses dedicated per-pad modulation state and a bounded Slicer Tape workspace. See page 3 for the current Speed voice policy.

FOURTH CUSTOM RELEASE / 2.3.4+C4

Install, recover and credits

Bitbox MkII only. The patcher always takes official firmware 2.3.4 as input.

Patch locally and install

1. **Back up the entire microSD card**, including projects, samples and presets. Download official 2.3.4 and extract the ZIP. At the browser patcher, select the original BITBOXMK2.bin, even when upgrading from +c, +c2 or +c3. Patching runs in your browser; the firmware is not uploaded. Input and output SHA-256 checks guard the patch.

2. Put the patched file in the card root, named exactly **BITBOXMK2.bin**. Remove duplicate-download suffixes such as (1). Power off, insert the card, then hold **INFO** while powering on. Release INFO when the update message appears.

3. Allow about 30 seconds. Keep power on and wait for automatic reboot. Check **2.3.4+c4** at Menu's lower-right or Tools' top-right. See the official MkII manual for the complete flashing procedure.

Official input: 674,896 bytes. SHA-256:
35ab7995ad0eca8484f3b28bc76389f10b2037dbb21a4a394e950d9329073e86

Customized +c4 output (777,704 bytes). SHA-256:
ce2a10e0e11a81ed8578bddfea8ad04c3c86263cf4c643075f382cb269c3431b

Return to official firmware

Replace the patched file on the card with the original unmodified **BITBOXMK2.bin** from the official ZIP. Repeat the INFO-held power-on update and wait for reboot. Confirm **2.3.4** in Menu. Keep backups of presets made before upgrading.

Reference and acknowledgment

First release (+c) video overview. The video demonstrates the original features; later-release controls are covered in this guide.

1010music created the firmware and hardware. vwal integrated the custom features and patcher. GEN, the modulation sources and several effects draw on bitbox-mk2-semi. Upstream work includes Portkit, Justin Joe’s blackbox-mod, Emilie Gillet’s stmlib and OpenTracker-derived GEN procedures. See third-party notices for attribution and license details.

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